

# K SHREEKRISHNA UPADHYAYA

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## PROFILE

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AI/ML engineering student. Fine-tuning transformer-based audio models for speech dysfluency detection, with additional experience directing an agentic, specification-driven development workflow on a separate project.

## EDUCATION

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### Bachelor of Engineering in Artificial Intelligence & Machine Learning

Vivekananda College of Engineering & Technology, Puttur (VTU, Belagavi)

Cumulative GPA (till 6th semester): 8.69/10.0

Relevant Coursework: Artificial Intelligence, Machine Learning, Data Structures

September 2023 – Present

DK, Karnataka

## PROJECTS

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### DADS (Detection and Analysis of Dysfluencies in Speech) | Python, PyTorch, Librosa

September 2025 – November 2025

Stuttering dysfluency detection system on the SEP-28k dataset, exploring both per-type and multi-label CNN architectures.

- Trained 5 binary CNN classifiers (one per dysfluency type: prolongation, block, sound repetition, word repetition, interjection) on mel-spectrogram features with per-type hyperparameter search across window size, hop length, and mel bands
- Explored a CNN-LSTM multi-label classifier as an alternative to separate per-type models
- Built a PyQt5 desktop application with real-time audio recording, playback, spectrogram visualization, and background stutter detection inference

The Major Project extends this work with a Wav2Vec2 + CNN-transformer hybrid architecture.

### FinDiary | Go, PostgreSQL, Flutter, gRPC

July 2026 – Present

Ongoing personal finance tracking application developed entirely through Specification-Driven Development — specifications are authored and refined before an agentic AI workflow generates implementation code, with all output reviewed for architectural consistency.

- Architecting Go backend services with PostgreSQL persistence and gRPC-based client-server communication
- Directing development of a Flutter cross-platform client, with offline-first local state management planned
- Practicing spec-first, AI-assisted implementation as a deliberate alternative to traditional hand-written development, alongside the fully hand-written systems work below

### Sn Project | C, x86\_64 Assembly

December 2025 – Present

Collection of 13 cross-platform C libraries (memory allocators, threading, logging, tracing) extracted from an experimental game engine into reusable, independently-versioned infrastructure.

- Designed 7 memory allocator strategies behind a unified interface; hand-wrote x86\_64 atomics with explicit memory-ordering fences for cross-platform threading primitives

### Snuk | C

April 2026 – Present

Embeddable scripting language with a custom lexer, recursive-descent parser, and tree-walking interpreter ( 6K lines of C), including reference-counted memory management.

### hash shell | C

January 2026 – March 2026

Open-source contributor with 17 merged pull requests, including a self-proposed multi-PR refactoring initiative to align the codebase with upstream coding standards.

### Additional Projects:

Regex Engine (Thompson NFA construction in C), StateFlow (DFA/NFA simulator with automaton validity checking), LC3VM (LC-3 virtual machine).

## SKILLS

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**AI/ML:** PyTorch, Transformers, Wav2Vec2, CNN Architectures, Model Fine-Tuning, Dataset Preprocessing & Evaluation, Specification-Driven / Agentic AI Workflows

**Languages & Build:** Python, C, C++, Go, Java, CMake, x86\_64 Assembly, POSIX Shell

**Development:** Git, Linux, CI/CD (GitHub Actions), gRPC, REST APIs, PostgreSQL, Flutter, Typst

**Achievements:** Winner of Nexathon 2025 (SDIT) and DevHack 2025 (Sahyadri College)